İSTANBUL TECHNICAL UNIVERSITY

FACULTY OF COMPUTER AND INFORMATICS

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA ENGINEERING

AI AND DATA ENGINEERING DESIGN

PROJECT CHARTER I

DESCRIPTION

The project will address the **critical inefficiency and lack of dedicated optimization in the deployment phase of trained Neural Networks (NNs)**. While advancements in the **training phase** (e.g., in architectures like Symbolic Regression, Neural ODEs, and Integral NNs) are paramount, the post-training deployment phase remains a secondary research focus, often relying on generalized compression techniques. This leads to deployed models that are computationally expensive for applications requiring high-volume, low-latency inference.

The core problem is to **create a systematic, automated methodology for optimizing a trained, basic NN specifically for fast and efficient deployment without sacrificing its prediction accuracy**. This approach focuses on the foundational model, as these principles are the basis for optimizing more complex architectures. The project will showcase how a compiler-like approach can convert a standard, trained model into a highly optimized, production-ready form.

This topic is highly relevant to **AI and Data Engineering** as it integrates deep learning concepts with practical system design and optimization. It is **multidisciplinary**, impacting fields such as **computer architecture** (hardware-agnostic optimization), **software engineering** (efficient code generation), and any domain utilizing large-scale AI deployment.

For the first part of this course, the following tasks will be completed:

- **Determining a complex engineering design problem:** Formally defining the trade-off between NN deployment efficiency and accuracy.
- **Conducting a comprehensive literature survey:** Researching model compression techniques (pruning, quantization, low-rank factorization) and dedicated neural network architectures that inherently support high-speed deployment.
- **Setting and analyzing design constraints:** Establishing clear, measurable

constraints on inference speed, memory footprint, and accuracy preservation.

- **Determining and analyzing relevant engineering standards:** Investigating deployment frameworks, model formats (e.g., ONNX), and industry-specific performance benchmarks.
- **Analyzing alternative solutions:** Evaluating different model compression and optimization techniques (e.g., **Knowledge Distillation** vs. **Quantization**) to select the most promising path.
- **Assessing relevant risks:** Identifying technical risks, such as loss of model accuracy, compatibility issues with deployment hardware, and the complexity of creating an automated optimization pipeline.
- **Developing a requirement analysis report:** Documenting the functional and non-functional requirements for the proposed optimized deployment system.
- **Developing the project work plan and schedule:** Creating a detailed plan for the execution phase of the project, including milestones for implementation and testing.

Ultimately, the objective is to demonstrate a methodology that can **automatically create a new, highly optimized deployment network** based on changes to the original training network, using a **common basic NN** as the proof-of-concept model.

1. PROJECT TEAM

ID	NAME	MAJOR
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2. PROJECT NAME

The Inference Compiler: Bridging Training and Deployment Phases of Neural Networks

3. SHORT SUMMARY (ABSTRACT)

Concept, Scope, and Application Area: This project addresses the critical efficiency gap in the post-training deployment of **Basic Neural Networks (NNs)** by designing an automated optimization methodology, using a standard NN as the foundational proof-of-concept. The findings are broadly applicable across fields requiring high-speed, high-volume inference, as basic NN principles underlie complex architectures.

Aim: The primary aim is to develop a systematic framework—**The Inference Compiler**—that takes a trained basic NN and transforms it into an equivalent, highly optimized model,

drastically increasing its inference speed and lowering computational cost while rigorously maintaining its original prediction accuracy.

Multi-Disciplinary Nature & Stakeholders: The project integrates principles from **AI and Data Engineering** (model compression, optimization algorithms), **Computer Architecture** (hardware-specific deployment), and **Software Engineering** (pipeline automation). Stakeholders include AI solution providers, software developers, and researchers seeking to deploy highly efficient foundational AI models.

Multiple Constraints and Design Criteria: Key constraints include the **maximum permissible drop in accuracy** (must be negligible), the **target inference speed increase** (e.g., 5X to 10X faster), and the **demonstration must use a basic NN architecture** to prove the foundational methodology. Design criteria prioritize **maintainability**, **automation**, and **accuracy preservation**.

Potential Measurable Objectives:

1. Achieve a minimum of 3X speedup in inference time on a target hardware platform for a chosen basic NN.
2. Quantify and limit the maximum drop in the benchmark model's accuracy to less than 2% of the original score.
3. Develop a fully automated script or tool that accepts the trained model and outputs the optimized deployment model.
4. Reduce the model's memory footprint by at least 70% through techniques like quantization or pruning.

Related Engineering Standards and their Relation to the Problem:

- **ONNX (Open Neural Network Exchange):** This standard will be central as it provides a common, portable format for moving models between different training frameworks and deployment runtimes, directly supporting the project's goal of creating a bridge between phases.
- **ISO/IEC 25010 (System and Software Quality Requirements and Evaluation):** The efficiency (performance) and maintainability characteristics defined in this standard will serve as the core metrics and constraints for evaluating the success of the optimized deployment model.

4. KEYWORDS

- Neural Networks (NN)
- Model Compression

- Inference Optimization
- Deployment Phase
- Basic Architectures
- Quantization
- Pruning
- Knowledge Distillation
- AI Engineering
- Computational Efficiency
- Automation

5. DESIGN CONSTRAINTS

Extensibility and Maintainability (The Automation Constraint): The entire optimization methodology, or "Inference Compiler," must be **extensible** and designed for **automation**. It must be maintainable enough so that if the basic NN's configuration is changed (e.g., number of layers, neurons), the pipeline can automatically adapt and generate an efficient deployment version, proving the system's ability to handle foundational changes.

Functionality and Accuracy Preservation (The Foundational Constraint): The optimized deployment model **must** preserve the **functionality** and **accuracy** of the original **Basic Neural Network (NN)** within a **negligible tolerance** (accuracy drop must be <= 2% of the original score). The success of the methodology relies on this fundamental demonstration, which can then be applied to more complex architectures.

Computational Cost and Performance (The Deployment Efficiency Constraint): The resulting deployment model **must** demonstrate a significant improvement in **performance** (3 times faster inference speed) and a reduction in **cost** (70% memory footprint) when run on a target deployment platform (e.g., a CPU, edge device, or specific GPU), compared to running the original, un-optimized basic NN. This establishes the clear value proposition of the "Inference Compiler."

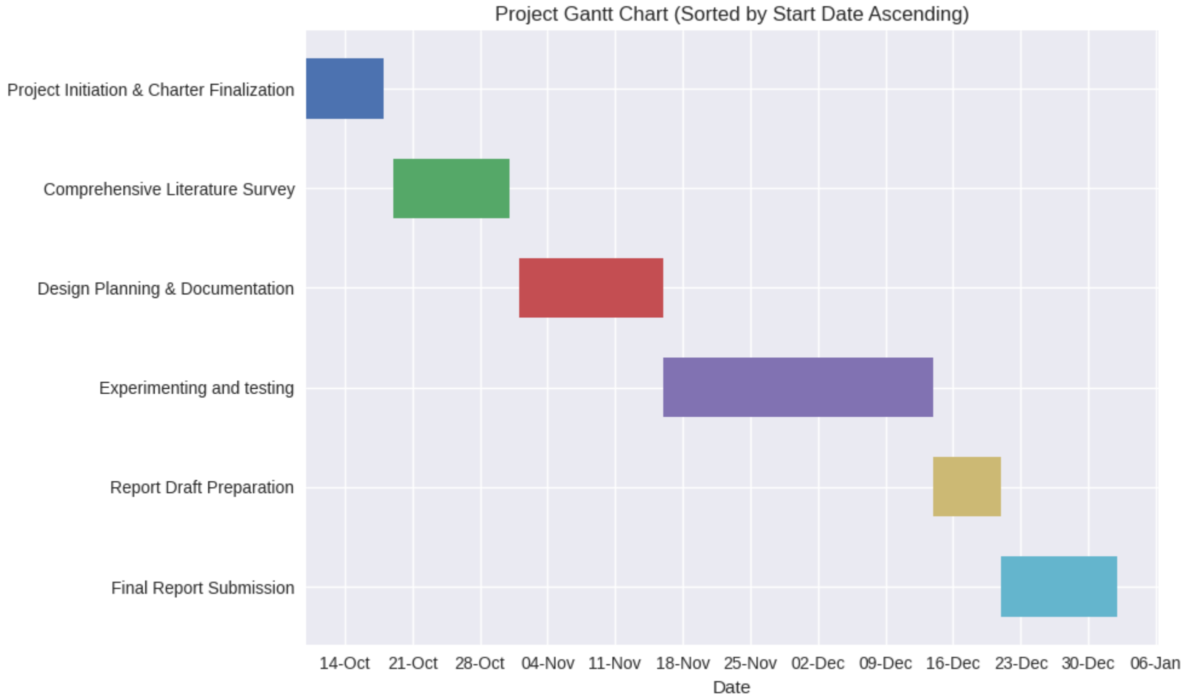
6. RELEVANT ENGINEERING STANDARDS

ONNX (Open Neural Network Exchange): ONNX is an open standard that defines a common set of operators and a file format to represent Deep Learning models. Relevent to the **Extensibility and Maintainability** constraint.
ISO/IEC 25010 (System and Software Quality Requirements and Evaluation): This international standard defines quality models for system and software products, categorizing quality characteristics like **Performance Efficiency** and **Maintainability**.

7. WORK BREAKDOWN, DELIVERABLES AND TIME PLAN

#	TASK	START DATE	DUE DATE	DELIVERABLE
1	Project Initiation & Charter Finalization	10-10-2025	18-10-2025	Project charter

2	Comprehensive Literature Survey	19-10-2025	31-10-2025	
3	Design Planning & Documentation	1-11-2025	16-11-2025	Interim report
4	Experimenting and testing	16-11-2025	14-12-2025	
5	Report Draft Preparation	14-12-2025	21-12-2025	Report Draft
6	Final Report Submission	21-12-2025	2-1-2026	Final report



8. Intellectual Property Rights

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9. SUPERVISOR INFO AND CONFIRMATION

Date of Proposal	18.10.2025
Academic Term of Project Delivery	**2025 Fall Term**
Student(s) Signature	
Project Advisor(s) Name	**Prof. Dr. Behçet Uğur TÖREYİN**
Project Advisor(s) Signature	